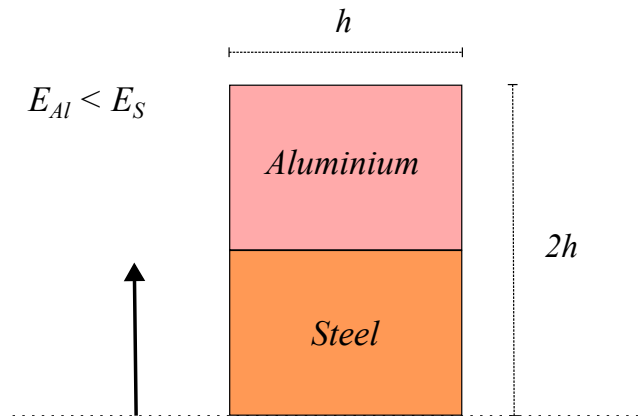


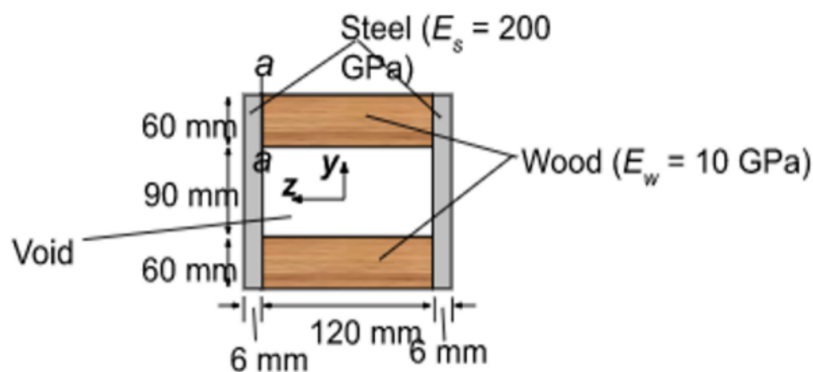
Tutorial 9

APL 108 - 2025 (Mechanics of Solids)

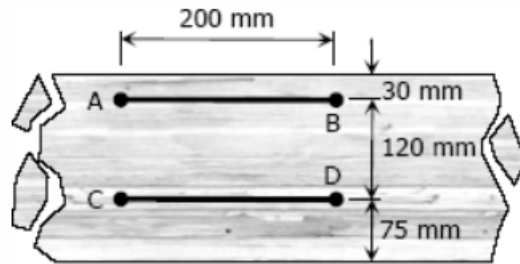
1. Think of a composite beam having a rectangular cross-section such that one half of the cross-section (having a square shape) is aluminum while the other half is steel. When such a beam is bent, where will the neutral axis lie in the cross-section (calculated from the bottom line of the cross-section)?



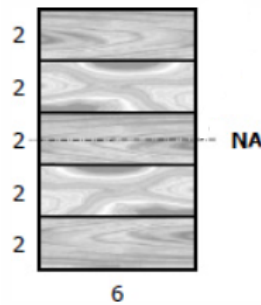
2. A beam of composite cross-section is subjected to bending moment $M_z = 30\text{kN}$. Find:
 - (a) The curvature induced in the beam
 - (b) Maximum bending stress in wood
 - (c) Maximum bending stress in steel



3. A flat steel bar, 2.5cm wide by 0.6cm thick and 1m long, is bent by couples applied at the ends so that the midpoint deflection is 2.51cm. Compute the stress in the bar and the magnitude of the couples. Use $E = 200\text{GPa}$.
4. In a laboratory test of a beam loaded by end couples, the fibers at layer AB in Fig are found to increase $60 \times 10^{-3}\text{mm}$ whereas those at CD decrease $100 \times 10^{-3}\text{mm}$ in the 200mm-gauge length. Using $E = 70\text{GPa}$, determine the flexural stress in the top and bottom fibers.



5. A laminated beam is composed of five planks, each 6 in. by 2 in. glued together to form a section 6 in. wide by 10 in. high.



The allowable shear stress in the glue is 90 psi, the allowable shear stress in the wood is 120 psi, and the allowable flexural stress in the wood is 1200 psi. Determine the maximum uniformly distributed load that can be carried by a simply supported beam on a 6ft simple span.

6. For an I-beam, assume the beam is subjected to transverse load
 - (a) Obtain an expression for variation in shear stress τ_{xy} within its cross-section. You can use the formula $\tau_{xy} = \frac{VQ(y)}{I_{zz}b(y)}$.
 - (b) Using the expression above, draw a graph depicting qualitative variation in shear stress within the cross-section.
 - (c) Where is the shear stress maximum? Find the ratio of maximum shear stress to average shear stress in the cross-section.

7. A cantilever beam (figure below) is made up of three wooden planks nailed together as shown. Find the expression for shear force in nails in terms of F , p , b , and h . For the cross-section: $I_{zz} = 10bh^3$.

